UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 4D – Physical Ergonomics: Occupational Biomechanics II

EIN3241: Human Factors and Ergonomics
Spring 2026

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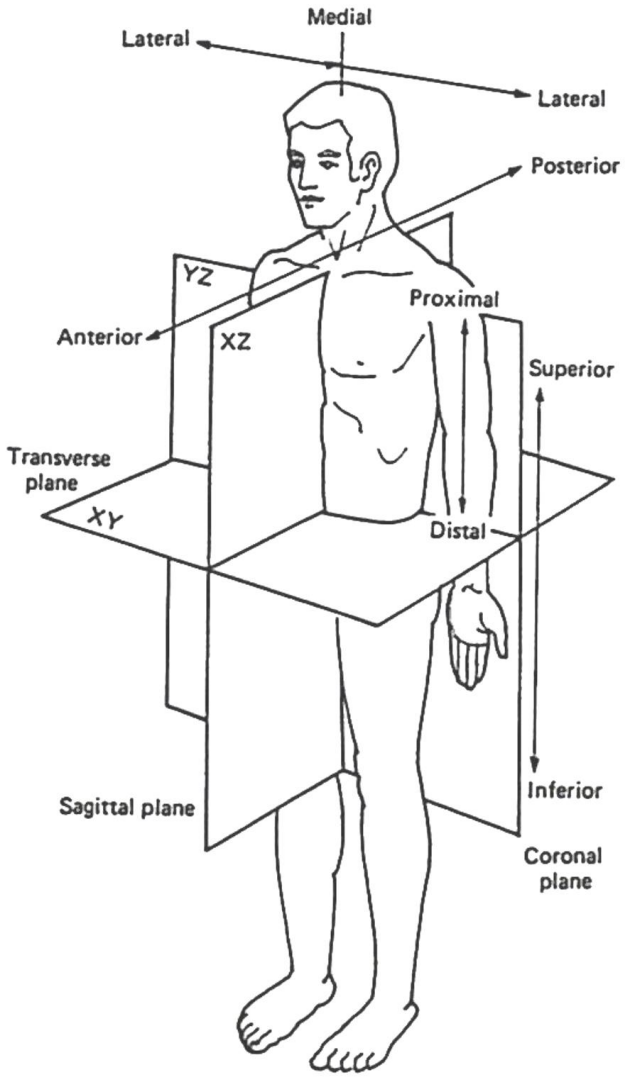
Admin Issues

- Quiz Wednesday
  - Lectures 4A, 4B, 4C, and "*4D*"
- Midterm March 9, 8:20-10:10 in FLG 270
- Extra Session for Biomechanics?

What did we talk so far?

- Biomechanical approach
- Two major branches of biomechanics
- Biomechanics model
- Body Planes (Sagittal, Frontal, Transverse)
- Anatomical Directions
  - Inferior/Superior
  - Anterior/Posterior
  - Medial/Lateral
  - Proximal/Distal
- Single Segment Model
- Segment-to-Segment Model
- Axis and/or force at angles…
- And more…



Biomechanics: Useful Tables

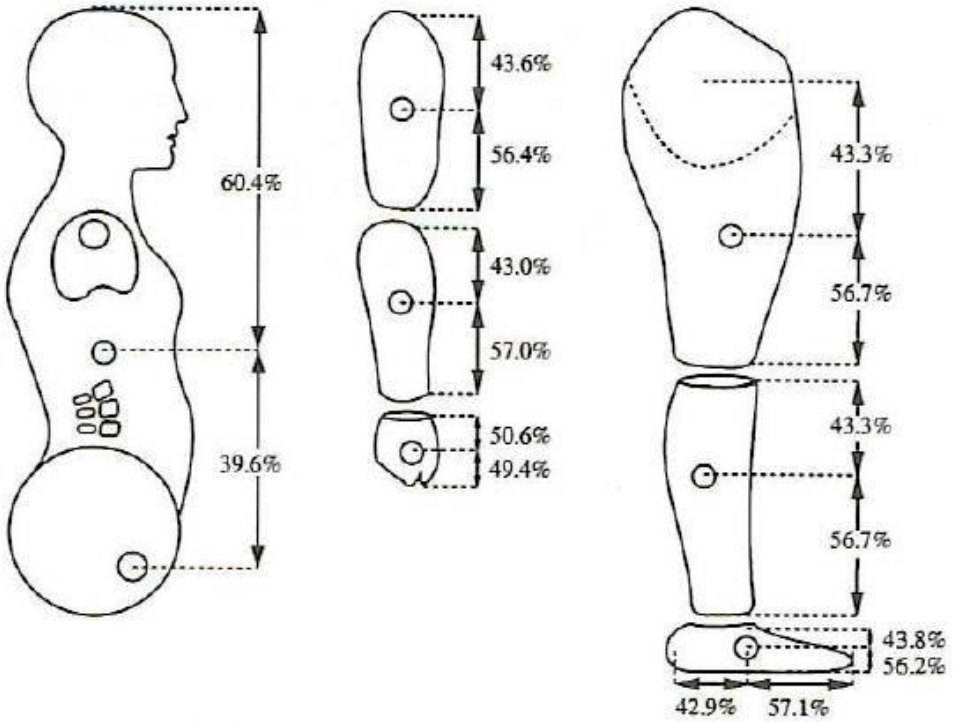


Fig. 4.3 The locations of centers of mass in the body segments in the sagittal plane indicated by percentages of the body segments. Adapted from Dempster (1955).

Table 4.4 Masses of body segments as a percentage of the whole body mass

Group body segments as a percentage of		Individual body segment mass as a percentage of		
	Total body mass (%)		Group segment mass (%)	Total body mass (%)
Head and neck	8.4	Head	73.8	6.2
		Neck	26.2	2.2
Torso (trunk)	50.0	Thorax	43.8	21.9
		Lumbar	29.4	14.7
		Pelvis	26.8	13.4
Each arm (total)	5.1	Upper arm	54.9	2.8
		Forearm	33.3	1.7
		Hand	11.8	0.6
Each leg (total)	15.7	Thigh	63.7	10.0
		Lower leg (shank)	27.4	4.3
		Foot	8.9	1.4

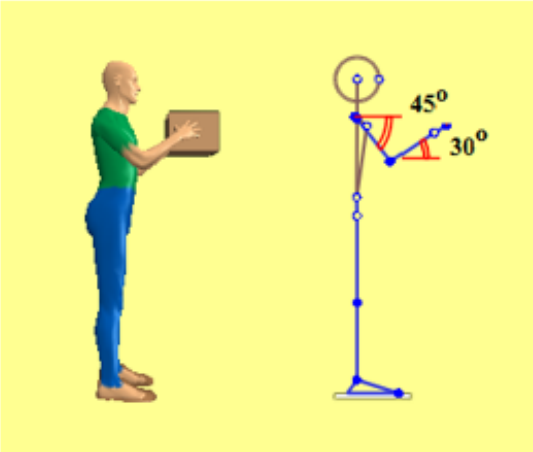
From: Tayyari and Smith (1997). Occupational Ergonomics Principals and Applications. Chapman & Hall: London, pp. 56-57.

Example

Consider the following information for the lifting task below:

- Mass of box = 15 kg
- Mass of person = 75 kg
- Hand length = 18 cm
- Forearm length = 30 cm
- Upper arm length = 34 cm
- Assume the weight of the box is applied at the center of gravity of the hands.

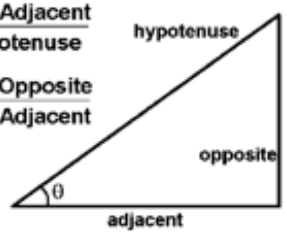
Determine the reaction moment and forces acting only on the right shoulder due to the upper extremity and weight being lifted.



$$\sin\theta = \frac{\text{Side Opposite}}{\text{Hypotenuse}}$$

$$\cos\theta = \frac{\text{Side Adjacent}}{\text{Hypotenuse}}$$

$$\tan\theta = \frac{\text{Side Opposite}}{\text{Side Adjacent}}$$

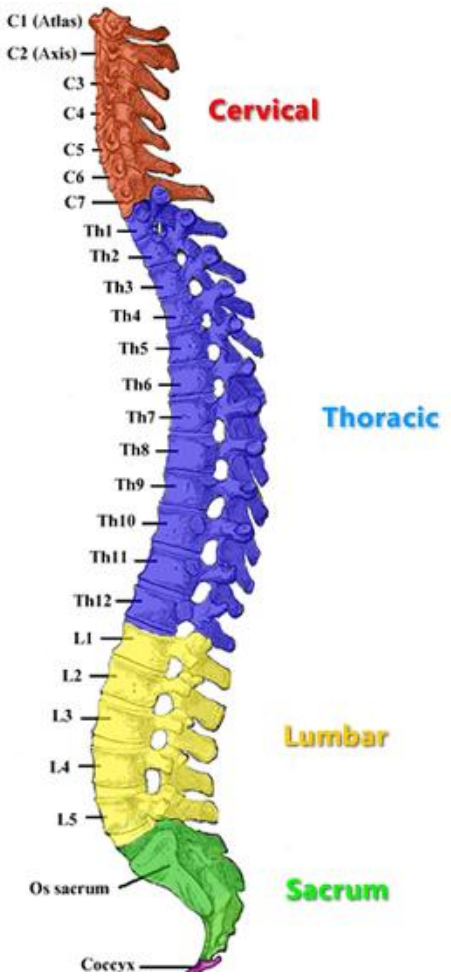


Solve in two approaches:

1) One Segment Approach
   - Which means the arm as one segment
2) Segment-to-Segment Approach
   - Which means you have to break the arm into segments by joints (wrist, elbow, and shoulder)
   - Along the way, prove that:
     - $R_E = 90.405\ N\ (upwards)$
     - $M_E = 27.78\ Nm\ (CCW)$

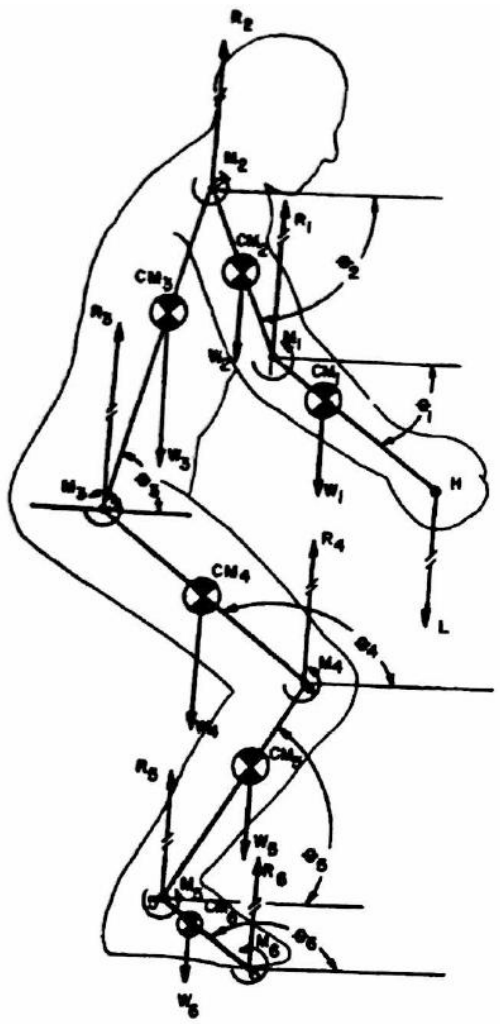
Back Models

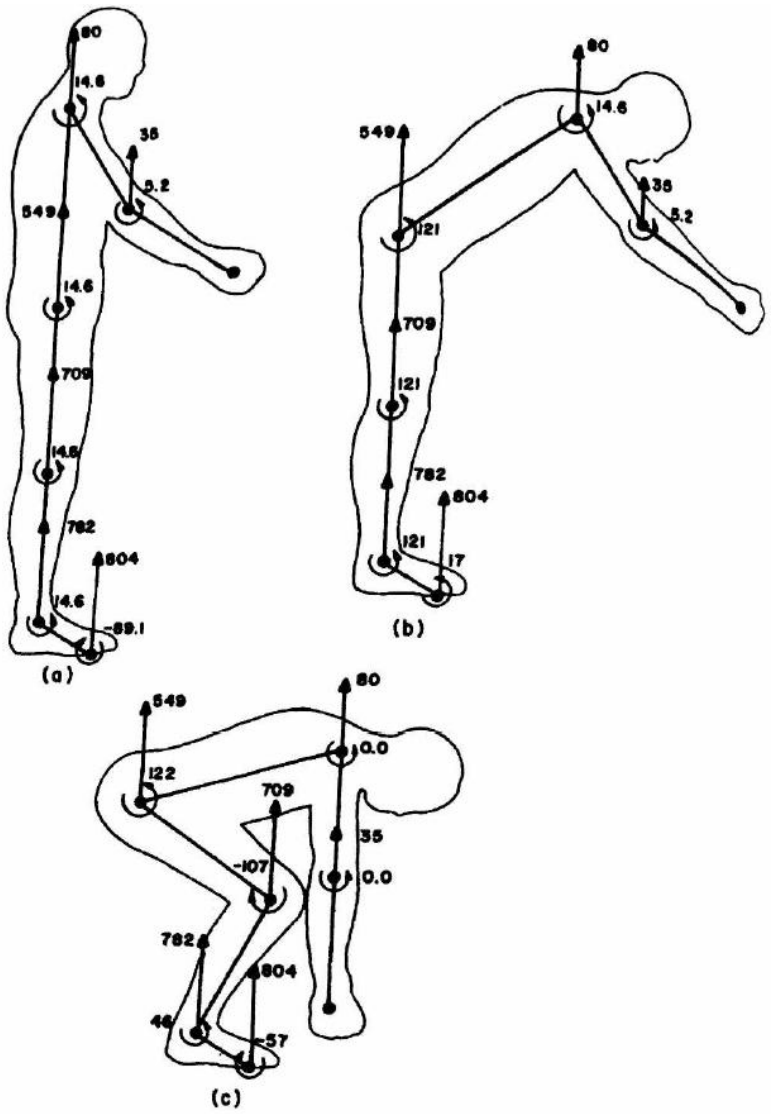
- The Vertebral Column has five major regions:
  - Cervical vertebrae
  - Thoracic vertebrae
  - Lumbar vertebrae
  - Sacrum
  - Coccyx



- The back typically experiences the most notable forces and moments
  - Why?
  - Especially which region of the back?

Back Models





From Biomechanics to Task Evaluation

- Compare job demands to human capacity.
  - Compare external moments to joint strength (maximum internal moment).
  - Use appropriate strength data (i.e. in same posture).

- Two types of evaluations:
  - Compare external moments with an individual's joint strength (e.g. screening).
  - Determine what percentage of the population is capable of performing the job.

- Ultimate question: Is the task or load acceptable?

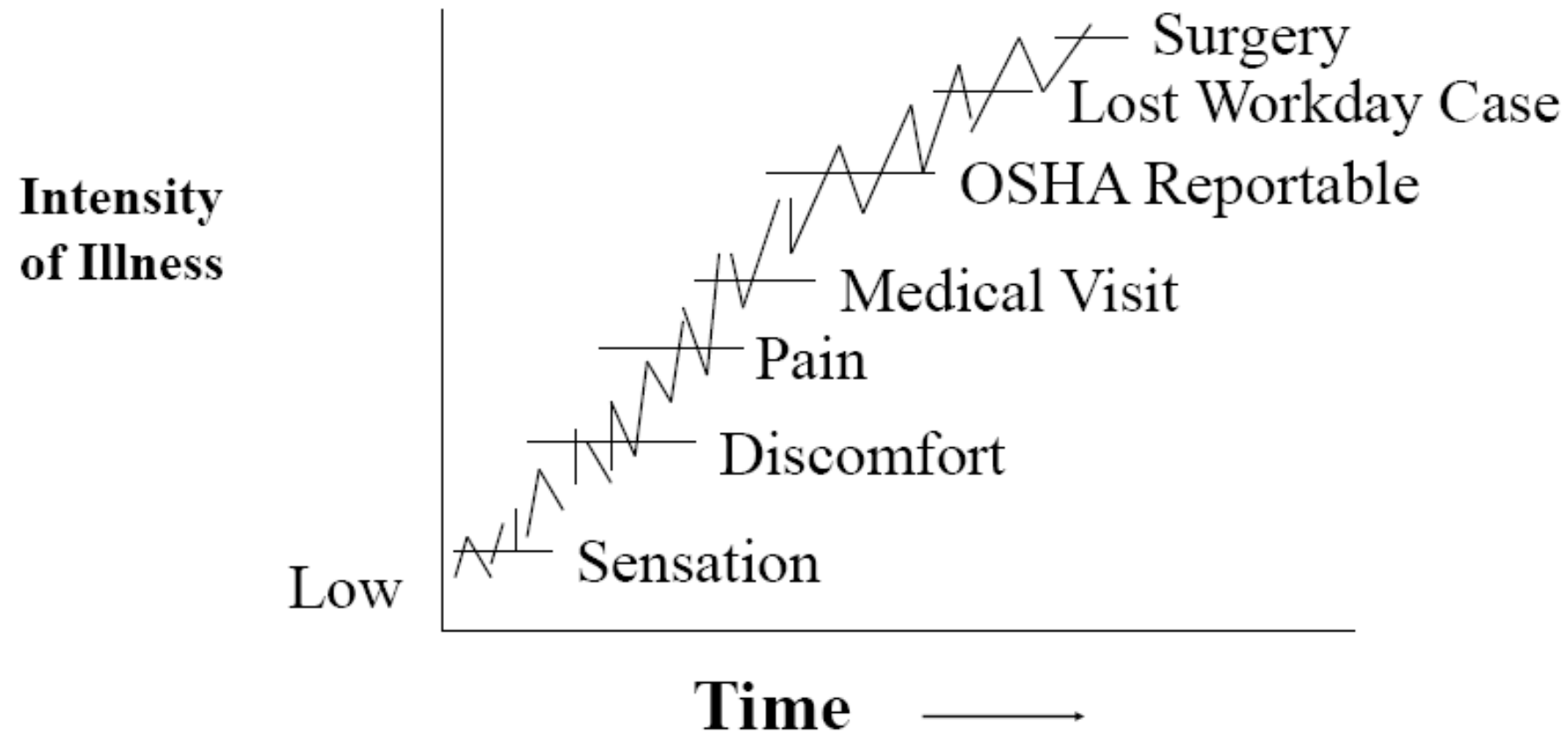
Unacceptable conditions lead to:

- **Work-Related Musculoskeletal Disorders (WMSD or MSD)** are pain or injuries in muscles, joints, ligaments, tendons, and/or nerves.
  - Most affected body parts are the back (46.5%) and shoulder (13.3%)

- Physical factors are the leading cause of WMSDs.

- What are other factors that lead to WMSDs?
  - Psychosocial
  - Personal
  - Environmental

Symptoms increase over time



Preventing WMSDs

- Institute Controls
  - Engineering Controls: changes made to the work environment, layout, tools, or equipment or changing the way a job is performed to avoid injuries.
  - Administrative Controls: Work practices and policies by the administration to reduce risk of injury.
  - Personal Protective Equipment (PPE): Equipment worn by the worker to reduce risk of injury.

Preventing WMSDs: Engineering Controls

- Examples:
  - Redesign the job to eliminate awkward postures
  - Allow for adjustability in the workstation and the worker (seating)
  - Locate tools within easy reach
  - Automation
  - Use fixtures and jigs to support work pieces
  - Round surfaces edges to avoid sharp protrusions
  - Design jobs to reduce hand forces and repetition
  - Utilize power grips whenever possible, use pinch grips only for precision tasks
  - Avoid over the shoulder and below knuckle height reaches

Preventing WMSDs: Administrative Controls

- Examples:
  - Job rotation
  - Warm-up/stretching programs
  - Alternate hands or balance the work across both hands
  - Control preexisting conditions
  - Work/rest cycle control
  - Avoid incentive pay scales
  - Train workers in correct postures and work methods and in early detection
  - Train management for early detection and control of new cases

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LEADING THE CHARGE, CHARGING AHEAD